Analytical AI in Sustainability Communication: Implications for Research and Education

Topic: ***The Adoption of AI in Strategic Communications*** **Subtopic:** ***The Future Impact of AI in Strategic Communications and PR Education***

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Abstract

Strategic communication is undergoing a profound transformation. While the rapid rise of generative AI—such as large language models for text creation or image generators for campaign visuals—has captured widespread attention, analytical forms of artificial intelligence provide equally important, and arguably more foundational, tools for empirical analysis and strategic planning. Analytical AI techniques—such as natural language processing (NLP) and computer vision—enable researchers and practitioners to examine large volumes of multimodal communication content in systematic and reproducible ways. This chapter argues that a more balanced understanding of AI adoption is needed: one that integrates the imaginative utility of generative models with the empirical rigor, scalability, and interpretive potential of analytical AI.

Sustainability Communication Research

Focusing on sustainability communication as a critical testbed, the chapter presents empirical findings from recent content analyses using transformer-based models such as BERT. For example, using semantic similarity metrics derived from sentence embeddings, we explore how key concepts—such as "energy" and "innovation" in Shell's messaging—are framed across campaign materials. This approach helps identify how corporations construct sustainability as a business opportunity, while NGOs like WWF may emphasize moral duty and community engagement. Such framing analyses offer communicators valuable insights into public positioning, value alignment, and stakeholder expectations.

Beyond semantic similarity, NLP supports more fine-grained analytical strategies. Aspect-based sentiment analysis, for instance, leverages part-of-speech tagging and syntactic parsing to isolate evaluative language tied to specific themes—such as “carbon emissions,” “renewable energy,” or “social justice.” This allows analysts to distinguish between overall sentiment and targeted evaluations, producing nuanced diagnostics of organizational messaging. Similarly, entity recognition and dependency parsing allow for the extraction of actor networks and the mapping of causal claims or value propositions across texts.

In the visual domain, computer vision expands the analytical toolkit for communication professionals. Object detection can identify recurring symbols, logos, or environmental imagery used in sustainability messaging. Facial expression analysis, using emotion detection models, enables researchers to interpret audience reactions in testimonial videos or campaign footage. Additionally, video summarization models can automate the identification of key scenes and shifts in visual tone, facilitating large-scale comparisons of audiovisual material in ways previously unfeasible.

Sustainability Communication Education

The chapter also considers the strategic implications of this shift. As communicative work becomes increasingly shaped by algorithmic processes, the role of the communication professional must evolve from message producer to message analyst. Analytical AI strengthens the strategist function by providing tools that support evidence-based decision-making, stakeholder analysis, and longitudinal benchmarking of communication efforts. This transformation requires an overhaul in communication education. Curricula must integrate AI literacy—not only in technical terms, but as a form of critical pedagogy that emphasizes interpretation, ethical reflection, and accountability. Transparency, reproducibility, and accessibility—core tenets of the open science movement—should underpin these educational efforts.

In sum, the future of AI in strategic communication lies not only in creative automation but in analytical interpretation. This chapter demonstrates how computational methods can be bridged with conceptual frameworks in sustainability communication, equipping both researchers and practitioners to navigate a complex and data-rich communicative landscape.

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Nils Holmberg holds a PhD in Media and Communication Science from Lund University (2016). His dissertation investigated how web advertising affects children aged 9–12 when solving online tasks, using experimental methods and eye-tracking to assess visual attention and comprehension. The study contributed to understanding the complex digital marketing ecosystem and the use of physiological methods in communication research. His current research focuses on computational content analysis for the social sciences, applying quantitative methods to extract and visualize content features from digital texts and images. This work explores how techniques like natural language processing and computer vision can help analyze communication patterns related to societal issues. Institutional website: https://portal.research.lu.se/en/persons/nils-holmberg

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Jörgen Eksell holds a PhD in Service Studies from Lund University (2013) and is a senior lecturer in Strategic Communication. His work focuses on sustainable development, investigating the interplay between human activities, sustainability, and resilience. In research, he has collaborated with various stakeholders to enhance policymaking and practice. In recent years, he has also done research on the professionalisation of strategic communication education. He has specialised expertise in research areas such as place branding, resilience, sustainability, value proposition, and communication work. Through his research, he aims to contribute to a more sustainable future by addressing complex environmental challenges and promoting effective communication strategies. His insights are vital for fostering environmental stewardship and advancing sustainability initiatives. Institutional website: https://portal.research.lu.se/sv/persons/jörgen-eksell